CHAPTER 9

DETERMINATION OF INCOME AND EMPLOYMENT

CHAPTER ESSENCE AT A GLANCE:
Master aggregate demand-aggregate supply equilibrium, consumption and savings functions, investment multiplier mechanism, and remedies for inflationary and deflationary gaps.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Components of Aggregate Demand:** In a two-sector economy, AD = C + I, where C is consumption function ($C = C_{bar} + bY$) with autonomous consumption and MPC, and I is autonomous investment.

• **Propensity Identities:** MPC + MPS = 1 and APC + APS = 1; as national income rises, APC declines while MPC remains relatively stable in the short run.

• **Two Equilibrium Approaches:** Macroeconomic equilibrium output is determined where Aggregate Demand equals Aggregate Supply (C + I = Y) or where Ex-Ante Savings equals Ex-Ante Investment (S = I).

• **Investment Multiplier Dynamics:** An initial change in investment expenditure generates multiple expansions in national income; Multiplier k = 1 divided by (1 - MPC) = 1 divided by MPS.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Consumption Function} \iff C = \bar{C} + bY \quad \text{where} \quad b = \text{MPC} = \frac{\Delta C}{\Delta Y}$$

Linear Keynesian Consumption Relationship

$$\text{Equilibrium Income Condition} \iff Y^* = C + I \iff S = I$$

AD-AS and Leakages-Injections Duality

$$\text{Investment Multiplier} \iff k = \frac{\Delta Y}{\Delta I} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

Magnification Factor of Autonomous Investment

$$\text{Fundamental Identities} \iff \text{MPC} + \text{MPS} = 1 \quad \text{and} \quad \text{APC} + \text{APS} = 1$$

Exhaustive Allocation of Incremental National Income

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse Deflationary Gap with Deficit Demand; Deficient demand is the shortfall in AD, while Deflationary Gap is the vertical distance by which AD falls short of aggregate supply at full employment.

• **Alert:** Multiplier k has a positive relationship with MPC and an inverse relationship with MPS; minimum value of k is 1 (when MPC=0) and maximum is infinity (when MPC=1).

• **Alert:** In equilibrium analysis, always distinguish between Ex-Ante (planned) and Ex-Post (actual realized) investment.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Marginal Propensity to Consume (MPC) and Marginal Propensity to Save (MPS). Prove that MPC + MPS = 1. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definitions:** MPC is the ratio of change in consumption to change in total income (MPC = Delta C divided by Delta Y); MPS is the ratio of change in savings to change in total income (MPS = Delta S divided by Delta Y). **[1 Mark]**

• **Proof:** We know that total income is either consumed or saved: Delta Y = Delta C + Delta S. Dividing both sides by Delta Y gives: 1 = (Delta C divided by Delta Y) + (Delta S divided by Delta Y) = MPC + MPS. **[1 Mark]**

2. Explain the determination of equilibrium level of income using the Saving-Investment (S-I) approach. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Equilibrium Condition:** In a two-sector economy, equilibrium output is attained where ex-ante planned savings equals ex-ante planned investment (S = I). **[1 Mark]**

• **Disequilibrium Adjustment when S > I:** When planned savings exceed planned investment, aggregate spending falls short of output. Unsold inventory stocks accumulate, forcing producers to cut output and employment until income falls and S drops back to equal I. **[1.5 Marks]**

• **Disequilibrium Adjustment when S < I:** When planned savings are less than planned investment, aggregate spending exceeds current output. Inventories deplete below desired levels, prompting producers to expand production and employment until income rises and S equals I. **[1.5 Marks]**

3. Explain the working of the Investment Multiplier with a numerical example assuming an initial investment increase of Rs 1,000 Crores and MPC = 0.8. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Formula:** k = 1 divided by (1 - MPC) = 1 divided by (1 - 0.8) = 1 divided by 0.2 = 5. Total increase in income Delta Y = k * Delta I = 5 * 1,000 = Rs 5,000 Crores. **[1 Mark]**

• **Round 1:** Additional investment of Rs 1,000 Crores generates income of Rs 1,000 Crores for capital goods producers. With MPC = 0.8, consumption increases by Rs 800 Crores and Rs 200 Crores is saved. **[1 Mark]**

• **Round 2:** Consumption spending of Rs 800 Crores becomes income for consumer goods producers. They spend 80 percent (Rs 640 Crores) on consumption and save Rs 160 Crores. **[1 Mark]**

• **Cumulative Total:** This multiplier process continues across successive rounds until total cumulative income reaches Rs 5,000 Crores and cumulative savings equal the initial investment of Rs 1,000 Crores. **[1 Mark]**

4. Distinguish between Inflationary Gap and Deflationary Gap. Explain one fiscal measure and one monetary measure to correct a Deflationary Gap. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Inflationary Gap vs Deflationary Gap:** Inflationary Gap is the excess of aggregate demand over full employment aggregate supply, creating demand-pull inflation. Deflationary Gap is the deficiency of aggregate demand below full employment aggregate supply, causing involuntary unemployment. **[2 Marks]**

• **Fiscal Measure to Correct Deflationary Gap:** Increase government public works expenditure and reduce income taxes to put more disposable purchasing power into households' hands. **[1 Mark]**

• **Monetary Measure to Correct Deflationary Gap:** Reduce Repo Rate and lower Cash Reserve Ratio (CRR), enabling commercial banks to expand credit at cheaper interest rates to boost private investment. **[1 Mark]**

5. In an economy, the consumption function is C = 100 + 0.75 Y and autonomous investment is I = Rs 500 Crores. Calculate: (a) Equilibrium level of income, (b) Consumption expenditure at equilibrium. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **(a) Equilibrium Income:** Condition is Y = C + I. Substitute: Y = 100 + 0.75 Y + 500. Y - 0.75 Y = 600. 0.25 Y = 600. Y = 600 divided by 0.25 = Rs 2,400 Crores. **[1.5 Marks]**

• **(b) Consumption at Equilibrium:** C = 100 + 0.75 * 2,400 = 100 + 1,800 = Rs 1,900 Crores. (Savings S = Y - C = 2,400 - 1,900 = Rs 500 Crores, matching Investment). **[1.5 Marks]**

6. What is the relationship between Marginal Propensity to Save (MPS) and the Investment Multiplier? **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **Inverse Mathematical Relationship:** Multiplier k = 1 divided by MPS. As MPS increases, households save more and consume less of incremental income, causing the multiplier effect to become smaller. **[1 Mark]**

• **Limiting Boundaries:** When MPS = 0, k is infinity; when MPS = 1, k reaches its minimum possible value of 1. **[1 Mark]**